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Tricho-Hepato-Enteric Syndrome mutation update: mutations spectrum of *TTC37* and *SKIV2L*, clinical analysis and future prospects

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Abstract

Tricho-Hepato-Enteric syndrome (THES) is a very rare autosomal recessive syndromic enteropathy caused by mutations of either *TTC37* or *SKIV2L* genes. Very little is known of these two gene products in mammals nor of the pathophysiology of the disease. Since the identification of the genes, we have set up the molecular diagnostic of THES in routine, gathering a large cohort with clinical and molecular data. Here, we report the phenotype and genotype analysis of this cohort together with an extensive literature review of THES cases worldwide, ie 96 individuals harboring mutations in one gene or the other. We set up locus-specific databases for both genes and reviewed the type of mutation as well as their localization in the proteins. No hot spot is evidenced for any type of mutation. The phenotypic analysis was first made on the whole cohort but is limited due to heterogeneity in clinical descriptions. We then examined the lab diagnostic cohort in detail for clinical manifestations. For the first time, we are able to suggest that patients lacking *SKIV2L* seem more severely affected than those lacking *TTC37*, in terms of liver damage and prenatal growth impairment.

Introduction

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Tricho-Hepato-Enteric syndrome (THES) or Syndromic Diarrhea (SD) is a rare autosomal recessive syndromic enteropathy for which molecular bases have only recently been described resulting in significant improvement of clinical description. It is caused by the mutation of either *TTC37* or *SKIV2L, 2* genes which encode two components of the human SKI complex [Hartley et al., 2010, Fabre et al., 2012]. It would seem that the first description of this syndrome was an observation by Stankler in the early eighties and is thus sometimes called Stankler syndrome [Stankler et al., 1982]. However, it was only later that it was clearly identified in two cohorts as an association of severe diarrhea with phenotypic anomalies and immune deficit, and renamed THES [Girault et al., 1994, Verloes et al., 1997]. Since then, several other names have been used by different teams like syndromic diarrhea or intractable diarrhea syndrome. The causative genes identification in 2010 and 2012 gave the opportunity to group all these denominations under a consensual and unique name which was used in the Gene Review for *TTC37* and *SKIV2L* [Fabre et al., 2018a]. With a global incidence estimated at 1/1.000.000 births, this syndrome is a very rare disorder and includes as its main features: intractable diarrhea at a very early onset, IUGR, characteristic hair abnormalities often with trichorrhexis nodosa, immune defects, liver disease and skin abnormalities [Fabre et al., 2014]. In this paper, we report 96 cases of THES from our experience and the literature with molecular and clinical data and performed analysis on clinical signs and complication frequencies to better describe the disease and genotype/phenotype correlation. First, we report the general analysis of the main clinical signs from previously published cases. Second, a more detailed analysis is reported from our in-house diagnostic cohort, as we set up an extensive data collection for these patients.

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The exosome and the Ski complex

The exosome is a multipartner protein complex involved in RNA degradation both by 3'-exonuclease and/or endonuclease activity. The RNA exosome is not only responsible for the mRNA degradation process, but also for other RNA quality control and maturation [Lebreton and Seraphin, 2008]. It has been described both in archeas and eukaryotic cells, prokaryotes only harboring a much simpler machinery. All components of the exosome are essential for cell viability [Mitchell et al., 1997]. The exosome co-localizes with important sites of RNA biogenesis, such as actively transcribed genes or the nucleolus, and is also engaged in RNA quality control [reviewed by Vanacova and Stefl, 2007].

The Ski complex (Fig. 1) is a tetraprotein complex including an RNA helicase subunit encoded by Ski2 in yeast or *SKIV2L* in humans, a subunit made of a protein Tetratrico Peptide Repeats (TRP) encoded by Ski3 in yeast or *TTC37* in humans and two subunits of a WD40 (Trp/Asp repeats) domains containing protein encoded by Ski8 or WDR61in humans. It has been well studied in yeast, where a crystal structure of its interactions with the exosome and RNA channeling has been obtained [Halbach et al., 2013]. The Ski complex, of which orthologues exist in plants, yeast and vertebrae, is a cytoplasmic cofactor of the RNA exosome and is involved in specific functions of the RNA exosome such as cytoplasmic mRNA degradation, particularly in case of STOP loss and/or gain and post-endoribo-nucleolytic clivage residues decay. The Ski complex is also known to play a role in antiviral responses [Schaeffer et al., 2010].

Three diseases have so far been associated with mutations of the exosome core proteins in humans: a pontocerebellar hypoplasia with hypomyelination, and a spinal muscular atrophy caused respectively by missense mutations in EXOSC3 or EXOSC8 [Wan et al., 2012, Boczonadi et al., 2014], and a syndrome associating early onset retinitis pigmentosa,

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progressive sensorineural hearing loss, hypothyroidism, short stature, brachydactyly, dysmorphism, premature ageing and mild intellectual disability caused by missense mutation of EXOSC2 [di Donato et al., 2016]. As these diseases do not overlap with the clinical phenotype of THE syndrome [Fabre et al., 2014], it is likely that the exosome and the Ski complexes, even if they share common functions, are also able to act separately in distinct cellular processes.

The genes

In humans, two genes have been shown to cause THES: *TTC37* [Hartley et al., 2010] and *SKIV2L* [Fabre et al., 2012].

TTC37 (OMIM#614589 – NM_014639.3 – NC_000005.9) is located on the long arm of chromosome 5 at position 5q15, and spans from base pair 95,463,895 to 95,555,007 on the reverse strand (GRCh38.p7). According to Ensembl, it has 14 transcripts, among them 3 are coding. The longest (ENST00000358746.6) comprises 42 exons and is 5706 nucleotides long, which is far over the mean transcript size in the human genome (1.3kb [Sudbery, 2002]). The first 3 exons are untranslated. The protein (Q6PGP7) comprises 1564 amino acids, contains twenty tetratricopeptide repeated (TPR) domains which are known to mediate protein-protein interactions and the assembly of multiprotein complexes.

The second gene causing THES, *SKIV2L* (OMIM#600478 – NM_006929.4 – NC_000006.11), is located on the short arm of chromosome 6 at position 21.33 (6p21.33), and spans from base pair 31,958,804 to 31,969,852 (GRCh38.p7). This gene has 16 transcripts, among them 3 are coding. The longest (ENST00000375394.6) comprises all 28 exons and is 4170 nucleotides long. The protein (Q15477) comprises 1246 amino acids,

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contains a Helicase superfamily- ATP-binding domain between AA 319 and 475 and is thought to be involved in exosome-mediated RNA decay as an external partner.

Variants

Variants are described according to current HVGS mutation nomenclature guidelines (http://www.hgvs.org/mutnomen/; [den Dunnen et al., 2016]), given that the A of the first coding codon (AUG) is numbered as the nucleotide +1 (Genbank accession numbers NM_014639 for *TTC37*, NM_006929 for *SKIV2L*). We use the HGVS-approved symbol * for premature termination codons. All variants are listed in Tables 2 and 3. A detailed list of all mutations by patient as well as clinical signs is given as Supplementary Table S1. All molecular diagnostics were performed by direct Sanger sequencing of *TTC37* and *SKIV2L* genes or whole exome NGS followed by confirmation by Sanger sequencing. For patients with only one mutation found, we performed CGH-array analysis in our cytogenetics laboratory for two of them in a routine context, but the gene regions are only poorly covered and not informative enough. Thus, we plan to sequence samples from these patients though whole genome sequencing to look for deep intronic or regulatory regions CNV. As regards the ACMG classification [Richards et al., 2015], all variants are in class 4 or 5, since most of them are non-sense mutations or splicing mutations and lead to a loss of function.

Types of mutations

In this paper, we analyzed a cohort of 96 patients from 85 different families, both from the literature (n=37) and from our own diagnostic lab recruitment (n=59). Among these patients, 47% were males, hence showing no sex-ratio bias. *TTC37* was mutated for 65 patients (68%), whereas *SKIV2L* was mutated for 31 others (32%) (Table 1). A small fraction of patients

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remains incompletely elucidated, since for 8/96 (8.3%), we identified only one mutation, either in one gene or the other. Patients come from all over the world, with a slight predominance of Mediterranean or Asian origin due to our region of recruitment (Fig. 2A).

The total number of mutated alleles is 184, and it (this?) comprises 86 different mutations, 58 for *TTC37* and 28 for *SKIV2L*. Altogether, our cohort (ie, excluding the patients reported by others) represents the majority of diagnosed patients (59/96 patients, Table 1).

For *TTC37*, which was the first gene reported as being associated with THES [Hartley et al., 2010] and involved in about 2/3 of the cases, the mutations are mainly nonsenses, inducing the creation of a premature STOP codon in the ORF (43%, 54/125 alleles, Fig. 2C). A significant number of events also induce splicing abnormalities (almost 25%, 31/125), by abolishing a consensus splice site, or by activating a cryptic site in one case [Hartley et al., 2010]. Frameshifts induced by point deletions or insertions are observed in 14% of the alleles (18/125) and missenses in 16% (20/125). In one case, a loss of START codon is predicted. Finally, one InDel event was found, replacing 3 nucleotides by a single C and inducing a frameshift. Out of the 65 *TTC37* mutated patients, 39 are homozygous carriers of the same mutated allele (60%), 21 are compound heterozygotes (32%) and 5 bear only one deleterious allele, the other mutational event remaining unknown (8%, Fig. 2B). As usual in a diagnostic context, the direct Sanger sequencing of the gene only comprises exons and intron boundaries. In such cases, maybe the second event affects a regulatory element or lies in an intron. Indeed, several of them are very long, especially in *TTC37*, with 23 introns (on 42) longer than 1 kb, and 10 of them longer than 3kb. In *SKIV2L*, only one intron is longer than 1.3kb.

Patients harboring mutations in SKIV2 are also mainly homozygotes (19/31, 63%), some are compound heterozygotes (8/31, 27%), and 3 remain with only one deleterious allele

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identified so far (10%), probably for the same reasons as for *TTC37* mutated patients (Fig. 2B and 2C). The distribution of the mutation types slightly differs from *TTC37*: the three main types of effects (frameshifts 25%, nonsenses 23%, missenses 19%) are observed almost in the same proportions, without the predominance of nonsense mutations. Mutations affecting splicing only represent 8% of the cases (5/59).

For both genes indeed, the proportion of missense mutations is lower than that commonly observed in autosomal recessive diseases [Zampieri et al., 2016, Froese et al., 2016 as examples].

To address the question of the impact of mutation type on the phenotype, for both genes, patients were split into different subgroups, according to the mutated gene (*SKIV2L* or *TTC37*) and the type of mutation (Loss of Function or Missense) they carry. If one allele led to a LoF and the other was a missense in the same patient, the patient was categorized in the missense group since the protein product only reflects the missense mutation. No significant difference was evidenced (data not shown).

Distribution of mutations

There is no obvious mutational hot spot in *SKIV2L*, with mutations spreading all along the sequence, even if the missense mutations seem to be preferentially located in the functional domains (Fig. 3A). Interestingly, a significant number of small in-frame deletions are found (26%, 15/59), all of them located in the C-terminus end of the protein, thus out of the functional helicase domain. Moreover, none of the nonsense, splice or frameshift mutations lie in this helicase domain.

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This bias in the mutational spectrum is not observed in *TTC37*, the mutations being observed all along the gene whatever their effect (Fig. 3B). There are two "hot" sites, with two recurrent mutations. One is the premature stop codon present in exon 28 (c.2808G>A) which is found 15 times in 8 different families, all originating from Asia (Pakistan, India). It may be interesting to note that at each side of this spot, 4 splicing events were characterized, as if the locus was prone to mutation (c.2779-2G>A and c.2921-2A>G). The other recurrent event is also a nonsense (c.4572G>A) located in exon 42. This event is found 17 times in 9 different Turkish families and led us to preferentially sequence exon 42 when receiving a sample from this origin. Exon 42 seems to be a mutational hot spot, since two other close missenses are found 4 and 3 times in the neighborhood (c.4507C>T and c.4514T>C respectively). We recently published a report on this hot spot, especially in a Turkish subgroup of patients all harboring a nonsense mutation in the last coding exon of *TTC37*, associated with a slightly milder form of the disease [Fabre et al., 2018b].

Databases

We created two different locus-specific databases with the Universal Mutation Database (UMD) tool, moderated by PB: http://www.umd.be/*SKIV2L*/ and http://www.umd.be/*TTC37*/. We recorded all previously published mutations after a careful search of the literature. We recorded phenotypes as well as geographic origin when data was available. We adjusted the nomenclature of certain variations in older reports to ascertain that base pair and protein sequence numbering were consistent and based on the same cDNA reference sequence (*TTC37*: NM_014639; *SKIV2L*: NM_006929). Reported variations were classified by type.

Clinical significance and detailed analysis of the reported cohort

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To date, THES is the only known Mendelian disorder associated with these two *TTC37* and *SKIV2L* genes. It is inherited as an autosomal recessive trait, with a penetrance of 100%. With the exception of a few patients with only one mutation evidenced, only strict homozygous or compound heterozygous cases have been reported, and digenism has never been observed.

The THE syndrome is generally considered to be a syndrome of neonatal enteropathy [Canani et al., 2015]. Particularly, it is characterized by the association of 9 main clinical signs: intractable diarrhea, hair abnormalities, facial dysmorphism, IUGR, immunodeficiency, skin abnormalities, liver disease, congenital cardiac defects and platelet anomalies. Statistical analysis, performed on the lab's cohort only because of more precise and homogenous clinical observations, showed that patients mutated in *SKIV2L* seem to be more severely affected than patients with *TTC37* mutations, with several signs being noticed earlier in life or more serious (see below). Both the global clinical description for the whole worldwide cohort, and a more detailed description for our in-house cohort gathering 59 cases, are presented below and illustrated in Fig. 4A and Tables 4 and 5. The detailed clinical analysis by patient is given as Supplementary Table S1.

Intractable diarrhea. This sign can be considered as constant, with 100% of clinically described patients affected, although data is lacking for one *TTC37* patient (out of 65) and 3 *SKIV2L* patients (out of 31). It usually begins in the first days of life but can be delayed up to 8 months, with no obvious link between the gene, the mutation type or the ethnic origin of patients. It is an intractable diarrhea which persists despite enteral rest, often described as watery and more rarely with blood. Pathology of the intestine is non-specific and most of the time described as normal.

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In our sub-cohort, the diarrhea leads to malabsorption and failure to thrive and required parenteral nutrition in 85% of patients. There is no statistical difference between patients mutated in either one gene or the other in the onset of parenteral nutrition (nor for mean or median age, data not shown).

Hair abnormalities are another nearly constant sign, concerning about 90% of clinically described patients. Hair is described as woolly, brittle and easily removable. In three quarters of the cases, when explored, trichorrhexis nododa is found. The diagnosis of this sign is not systematically explored probably because it has no clinical consequence.

As regard to the "wooly hair" or "easily removable" criteria, it is present in about 87% of patients with no obvious difference whether one gene or the other is mutated (Table 5).

Facial dysmorphism is also nearly constant and present in 84% of clinically described patients. The main signs are a large forehead with broad nasal root and hypertelorism, but dysmorphism can become more apparent with time. However, this sign is often poorly described in detail in papers, with only the mention of "presence" or "absence" of dysmorphism. In the in-house cohort, even if we collected more details, there is no obvious correlation between the genes and the signs (Table 5).

IUGR/SGA is very frequent, observed in 70% of the *TTC37* mutated and 86% of the *SKIV2L* mutated patients clinically explored. Most children are under the 10th percentile at birth, either for height or weight, as previously described in the literature.

To deeply explore the growth of patients, we converted the weights and heights at birth in percentile for 59 patients and performed a statistical comparative analysis (Fig. 4B and 4C). At first glance, we confirm that patients are much smaller and lighter than healthy patients at birth (global comparative analysis, patients categorized by percentile fractions corresponding

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to multiple of standard deviations and compared to normal slots). Interestingly, a significant difference can be evidenced depending on the mutated gene (Table 4). For height, the mean percentile at birth is 19.89 (±29.25) for *TTC37*, whereas it is 2.20 (±4.46) for *SKIV2L* mutated patients. Thus, *SKIV2L* mutated patients seem to be smaller at birth when compared to *TTC37* ones (p-value: 0.01, t-test). For weight, the mean percentile at birth is 17.24 (±26.32) for *TTC37*, whereas it is 6.09 (±12.85) for *SKIV2L* mutated patients. Thus again, *SKIV2L* mutated patients seem to be lighter at birth when compared to *TTC37* ones (p-value: 0.04, t-test), and this last data is of stronger significance since the percentage of described patients is higher for weight than for height.

Liver disease is very frequent and is more frequent in patients with mutations in *SKIV2L* (88%) than in *TTC37* (51%). The reported abnormalities range from fibrosis and cirrhosis to elevated liver enzymes and hepatomegaly. The pathology showed iron overload and sometimes hemochromatosis.

A tendency seems to emerge, even if not statistically significant: in about 70% of the patients deeply analyzed, an initial cholestasis is reported for 40% (6/15) of *SKIV2L*, but only for 12% (3/25) of *TTC37* mutated patients (p-value: 0.057, Xhi2 test). The fact is similar with initial liver disease (53% vs 24% respectively) or hepatomegaly (43% vs 17%), and indeed for all initial signs explored. After parenteral nutrition (85% of described patients), these proportions become very similar between the two subgroups (for instance, 22% of *SKIV2L* and 21% of *TTC37* mutated patients present a secondary cholestasis) due to the effect of PN itself (Table 5).

Immunodeficiency and platelet anomalies. This sign is not very well delineated and is reported in about half of clinically explored patients. Patients often present a low

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immunoglobulin count or lack of antibody response to vaccinal immunization. Monoclonal hyper-IgA has been described.

In the sub-cohort, 73% of the patients were explored for this sign, and among them, 53.5% show a “low immunoglobulin level”. Interestingly, out of the 40/59 (68%) patients analyzed, the percentage of those showing platelet shape anomalies is 14% if they have a *TTC37* mutation, and 33% if they have a *SKIV2L* mutation, although not significant (p-value: 0.21, Xhi2 test).

Skin abnormalities. Between 39 and 53% of the children present dermatological abnormalities depending on the mutated gene, even if the percentage of described patients is relatively modest for this sign (less than 60%). The most frequently observed sign is “café au lait” spots, but sometimes also rubbery skin or xerosis. Café au lait spots are preferentially located in the lower limbs [Monnies et al., 2015].

In the sub-cohort, 51% of the patients do have skin anomalies, 30% present café au lait spots (15/49) and only 10% show xerosis (5/49). Other depigmented spots have been described, as well as in one case an angioma or in another, an ectodermal dysplasia.

Congenital cardiac defects are a less frequent sign, reported in about one third of clinically described patients. Children can present cardiac abnormalities ranging from Fallot disease to peripheral pulmonary stenosis. The most common sign observed is an atrial or ventricular septal.

Out of 46 patients further explored for cardiac function, only 9 are reported to have a cardiac defect (20%). Among them, 2 ventricular septal defects, 2 atrial septal defects, 3 interventricular communications and 2 interatrial communications were described.

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Mild mental retardation is reported in about half of the patients (Table 5), but this sign is usually not well documented as most of the children are very young, and we can't rule out an effect of the necessity of parenteral nutrition and hospital stays taking precedence over the school syllabus.

Growth

For about one third of patients during normal care, we have collected some data on their growth. It appears that the growth of either *SKIV2L* or *TTC37* mutated patients is severely impaired: Z-scores are -2.64±2.09 SD for height, and -2.67±2.57 SD for weight. Subsequently, the Body Mass Index is also under normal values, at -0.84±1.59 SD, but there is no significant difference between two subgroups. It is rather difficult to link that fact to the special nutritional regimen of patients, as a lot of them include special dietary requirements in their care [Fabre & al., 2017] (Z-scores were calculated with the help of the http://www.quesgen.com/BMIPedsCalc.php algorithm, based on the Atlanta Center for Disease Control, The USA: https://www.cdc.gov/growthcharts/percentile_data_files.htm, complete data not shown). Due to the small size of the groups, it is also difficult to link these data with parenteral/normal nutrition, or with any other parameter.

Intra-familial variability

The number of families is relatively small in the whole cohort (5 for *TTC37*, 3 for *SKIV2L*). To address the issue of whether affected siblings phenotypes are fully comparable, we compared all the data available. Even if we do find some signs in common within siblings, no family shows a perfect phenotypic homogeneity. All these data are available in Supplementary Table S1.

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Discussion

This paper is an attempt to report a detailed analysis of all the THES cases known worldwide. Since the description of the causative genes for THES, significant advances have been made in the clinical description and the understanding of this syndrome. The clinical features are now precisely delineated, and the molecular basis seems limited to the 2 genes described up to now. These genes, *TTC37* for a third of patients, and *SKIV2L* for the remaining third, encode for 2 of the subunits of the SKI complex, a multipartner proteic complex associated to the cytoplasmic exosome and involved in RNA processing and degradation.

The fact that about 60% of patients are mutated in *TTC37* is not that surprising, considering that this gene spans along a much longer chromosomic region than *SKIV2L* (91 kb vs 10 kb), and that the protein product is 300 amino acids longer. Loss of function in any of these genes leads to remarkably similar clinical signs, with the same core group of organs affected: mainly gut, hair and face and patients with either *SKIV2L* or *TTC37* mutations are indistinguishable in clinical practice. Such a functional overlap is rather common in recessive genetic diseases and is thus not surprising. For instance, muscular dystrophies LGMDII, due to defects in sarcoglycan proteins, share very similar distal muscle defects [Pegorado and Hoffman, 2012], or Walker-Warburg syndrome, glycosylation anomalies due to defects in several enzymes [Vajsar and Schachter, 2006]. It is thus likely that the symptoms of this severe syndrome are the consequence of the functional inactivation of the cytosolic RNA exosome in general RNA turnover.

However, we show here for the first time that some differences exist regarding clinical severity, as patients mutated in *SKIV2L* are affected more severely and/or earlier. The liver disease is more severe when *SKIV2L* is involved, and also, height and weight at birth are significantly smaller. The reason for that remains largely unknown, and needs to be confirmed in the future. One can hypothesize that *SKIV2L* has another function independent

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from the SKI complex at cellular level, as suggested by Eckard and collaborators. Notably, it was shown that human cells with *SKIV2L* deficiency, but not with *TTC37* deficiency, have a strong type I IFN signature, indicative of a chronic antiviral response. This is mediated by *SKIV2L* which has a unique role of negative regulator of Rig-like receptors, receptors which detect RNA and are the support of cell-intrinsic innate antiviral responses [Eckard et al., 2014]. This would suggest that THES patients with mutations in *SKIV2L* could be prone to develop autoimmune diseases, a fact never observed up to now, maybe due to the young age and shortened lifespan of these patients.

It should also be noted that a SNP (rs429608) in *SKIV2L* has been associated with a protective effect against Age-related Macular Degeneration [Kopplin et al., 2010]. These sparse data corroborate the fact that *SKIV2L* could have functions that are independent from *TTC37* and the exosome and that mutations in these genes can lead to some different functional defects, but again in an age-related pathology that cannot be observed in THES patients. Not only may *SKIV2L* act in a SKI complex independent manner, but that side effect could also imply another partner, such a non-coding RNA, or maybe that *SKIV2L* comprises an uncharacterized regulatory subunit. Also, we recently published a study on a Scandinavian sub cohort of patients mainly from Turkish origin, all harboring a C-terminal nonsense mutation of *TTC37* (all patients with the c.4572G>A homozygous allele in Table 2). Interestingly, these patients present a slightly milder phenotype than the other *TTC37* mutated ones, suggesting that a premature STOP codon in the last coding exon of the gene is less deleterious than others probably because the part of the protein is functionally dispensable [Fabre et al., 2018b].

No mutation in WDR61, the 3rd partner of the Ski complex has been reported so far in THES patients although it has been actively searched. This 3rd partner is the smallest of the SKI complex in humans and is present as a dimer. Other roles of this small protein have been

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documented, as WDR61 is part of another complex, the PAF complex, which is involved in transcriptional regulation and histone modifications [Zhu et al., 2005]. These other functions are probably the reason why alterations of the WDR61 protein do not result in THES phenotype.

The exosome and SKI complex are physically linked by a small isoform of HBS1L, HBS1LV3 (Ski7 in yeast) [Kalisiak et al., 2017]. Terminal mutations of HBS1L have been described in a syndrome associating severe intrauterine growth restriction with several other malformations, mainly of the urinary tractus and of the face. However, it seems it only affects HBS1LV in another role independent from the linker between the exosome and Ski complex, and no overlap can be evidenced with THES [Sankaran et al., 2013].

Up to now, mutations in 3 other exosome proteins have been shown to cause monogenic diseases: EXOSC2, EXOSC3 and EXOSC8, but only sparse characteristics are shared with THE, reinforcing the hypothesis of independent functions of the SKI complex [Rudnik-Schöneborn et al., 2013; Wan et al., 2012; Di Donato & al., 2016]. Another cofactor (RBM7) of the nuclear exosome targeting complex (NEXT) has recently been reported to be involved in a human disease, also with neural damages (SMA-like phenotype) [Giunta et al., 2016]. It is likely that these neurological diseases are rather due to the core function of the exosome in the nucleus compartment.

Despite the progress in clinical and molecular understanding of how the alteration of a basic function such as RNA decay can precisely lead to specific signs such as hair abnormalities or intractable diarrhea remains an unsolved question. For the future, a better understanding of the links between the molecular defects of the SKI complex and clinical signs will be crucial to the improvement of patient care as well as to the development of our knowledge of ARN turn-over in humans. Several hypotheses can be made to link protein alterations and clinical expression, including the persistence of abnormal RNAs that would affect some cell

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functions because of the accumulation of abnormal translation products due to the defect of the SKI complex. For mutations that alter RNA sequence, one can also hypothesize that an abnormal *SKIV2L* or *TTC37* RNA product exerts a toxic function in the cytoplasm, even if it seems unlikely. Another hypothesis could be the alteration of antiviral response or the alteration of microbiota/host relations. To this aim, animal model developments will be mandatory in the next years. Indeed, most of our current knowledge on the Ski complex comes from yeast experiments and even if there is a strong conservation of the Ski complex and associated exosome among eukaryotes, there are some significant differences between yeast and mammals in the exosome structure and functions. Given the fact that THES includes some developmental features such as IUGR, intellectual deficiency and dimorphism, drosophila is a suitable model that we are currently developing in a collaborative project.

Up to now, the clinical management of THES has generally consisted of the treatment of specific manifestations and the screening of complications such as severe liver disease [Fabre et al., 2017]. It is our hope that specific therapeutic approaches will result from animal model studies and especially from the advances in understanding about the role of environment interactions in the physiopathology of the disease.

Ethical disclosure

All patients (and or legal representatives) gave written informed consent for molecular analysis, storage of clinical information and DNA samples, according to the Helsinki protocol and rules.

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References

Aly HH, Suzuki J, Watashi K, Chayama K, Hoshino S, Hijikata M, Kato T, Wakita T. 2016. RNA Exosome Complex Regulates Stability of the Hepatitis B Virus X-mRNA Transcript in

This article is protected by copyright. All rights reserved.

Accepted Article

a Non-stop-mediated (NSD) RNA Quality Control Mechanism. J Biol Chem 291(31):15958-15974.

Bick D, Fraser PC, Gutzeit MF, Harris JM, Hambuch TM, Helbling DC, Jacob HJ, Kersten JN, Leuthner SR, May T, North PE, Prisco SZ, Schuler BA, Shimoyama M, Strong KA, Van Why SK, Veith R, Verbsky J, Weborg AM Jr, Wilk BM, Willoughby RE Jr, Worthey EA, Dimmock DP. 2017. Successful Application of Whole Genome Sequencing in a Medical Genetics Clinic. J Pediatr Genet 6(2):61-76.

Boczonadi V, Müller JS, Pyle A, Munkley J, Dor T, Quartararo J, Ferrero I, Karcagi V, Giunta M, Polvikoski T, Birchall D, Princzinger A, Cinnamon Y, Lützkendorf S, Piko H, Reza M, Florez L, Santibanez-Koref M, Griffin H, Schuelke M, Elpeleg O, Kalaydjieva L, Lochmüller H, Elliott DJ, Chinnery PF, Edvardson S, Horvath R. 2014. EXOSC8 mutations alter mRNA metabolism and cause hypomyelination with spinal muscular atrophy and cerebellar hypoplasia. Nat Commun 5:4287.

Bozzetti V, Bovo G, Vanzati A, Roggero P, Tagliabue PE. 2013. A new genetic mutation in a patient with syndromic diarrhea and hepatoblastoma. J Pediatr Gastroenterol Nutr 57:e15.

Busoni VB, Lemale J, Dubern B, Frangi F, Bourgeois P, Orsi M, Badens C, Fabre A. 2016. IBD-Like Features in Syndromic Diarrhea/Trichohepatoenteric Syndrome. J Pediatr Gastroenterol Nutr 64(1):37-41.

Canani RB, Castaldo G, Bacchetta R, Martín MG, Goulet O. 2015. Congenital diarrhoeal disorders: advances in this evolving web of inherited enteropathies. Nat Rev Gastroenterol Hepatol 12(5):293-302.

This article is protected by copyright. All rights reserved.

Accepted Article

Chong JH, Jamuar SS, Ong C, Thoon KC, Tan ES, Lai A, Aan MK, Tan WL, Foo R, Tan EC, Lau YL, Liew WK. 2015. Tricho-hepato-enteric syndrome (THE-S): two cases and review of the literature. Eur J Pediatr 174:1405-1411.

den Dunnen JT, Dalgleish R, Maglott DR, Hart RK, Greenblatt MS, McGowan-Jordan J, Roux AF, Smith T, Antonarakis SE, Taschner PE. 2016. HGVS Recommendations for the Description of Sequence Variants: 2016 Update. Hum Mutat 37(6):564-569.

Di Donato N, Neuhann T, Kahlert AK, Klink B, Hackmann K, Neuhann I, Novotna B, Schallner J, Krause C, Glass IA, Parnell SE, Benet-Pages A, Nissen AM, Berger W, Altmüller J, Thiele H, Weber BH, Schrock E, Dobyns WB, Bier A, Rump A. 2016. Mutations in EXOSC2 are associated with a novel syndrome characterised by retinitis pigmentosa, progressive hearing loss, premature ageing, short stature, mild intellectual disability and distinctive gestalt. J Med Genet 53(6):419-425.

Eckard SC, Rice GI, Fabre A, Badens C, Gray EE, Hartley JL, Crow YJ, Stetson DB. 2014. The *SKIV2L* RNA exosome limits activation of the RIG-I-like receptors. Nat Immunol 15(9):839-845.

Fabre A, Bourgeois P, Chaix C, Bertaux K, Goulet O, Badens C. 2018a. Trichohepatoenteric Syndrome. In: Adam MP, Ardinger HH, Pagon RA, et al., editors. GeneReviews® [Internet].

Fabre A, Petit LM, Hansen LF, Wewer AV, Esteve C, Chaix C, Bourgeois P, Badens C, Paerregaard A. 2018b. A new mutation in the C-terminal end of *TTC37* leading to a mild form of syndromic diarrhea/tricho-hepato-enteric syndrome in seven patients from two families. Am J Med Genet A. 2018 Jan 31.

This article is protected by copyright. All rights reserved.

Accepted Article

Fabre A, Bourgeois P, Coste ME, Roman C, Barlogis V, Badens C. 2017. Management of syndromic diarrhea/tricho-hepato-enteric syndrome: A review of the literature. Intractable Rare Dis Res 6(3):152-157.

Fabre A, Badens C. 2014. Human Mendelian diseases related to abnormalities of the RNA exosome or its cofactors. Intractable Rare Dis Res 3(1):8-11.

Fabre A, Breton A, Coste ME, Colomb V, Dubern B, Lachaux A, Lemale J, Mancini J, Marinier E, Martinez-Vinson C, Peretti N, Perry A, Roquelaure B, Venaille A, Sarles J, Goulet O, Badens C. 2014. Syndromic (phenotypic) diarrhoea of infancy/tricho-hepato-enteric syndrome. Arch Dis Child 99(1):35-38.

Fabre A, Charroux B, Martinez-Vinson C, Roquelaure B, Odul E, Sayar E, Smith H, Colomb V, Andre N, Hugot JP, Goulet O, Lacoste C, Sarles J, Royet J, Levy N, Badens C. 2012. *SKIV2L* mutations cause syndromic diarrhea, or trichohepatoenteric syndrome. Am J Hum Genet 90:689-692.

Fabre A, Martinez-Vinson C, Roquelaure B, Missirian C, André N, Breton A, Lachaux A, Odul E, Colomb V, Lemale J, Cézard JP, Goulet O, Sarles J, Levy N, Badens C. 2011. Novel mutations in *TTC37* associated with tricho-hepato-enteric syndrome. Hum Mutat 32:277-281.

Fabre A, André N, Breton A, Broué P, Badens C, Roquelaure B. 2007. Intractable diarrhea with "phenotypic anomalies" and tricho-hepato-enteric syndrome: two names for the same disorder. Am J Med Genet 143A(6):584-588.

Froese DS, Huemer M, Suormala T, Burda P, Coelho D, Guéant JL, Landolt MA, Kožich V, Fowler B, Baumgartner MR. 2016. Mutation Update and Review of Severe Methylenetetrahydrofolate Reductase Deficiency. Hum Mutat 37(5):427-438.

This article is protected by copyright. All rights reserved.

Accepted Article

Girault D, Goulet O, Le Deist F, Brousse N, Colomb V, Césarini JP, de Potter S, Canioni D, Griscelli C, Fischer A, et al. 1994. Intractable infant diarrhea associated with phenotypic abnormalities and immunodeficiency. J Pediatr 125(1):36-42.

Giunta M, Edvardson S, Xu Y, Schuelke M, Gomez-Duran A, Boczonadi V, Elpeleg O, Müller JS, Horvath R. 2016. Altered RNA metabolism due to a homozygous RBM7 mutation in a patient with spinal motor neuropathy. Hum Mol Genet 25(14):2985-2996.

Halbach F, Reichelt P, Rode M, Conti E. 2013. The yeast ski complex: crystal structure and RNA channeling to the exosome complex. Cell 154(4):814-826.

Hartley JL, Zachos NC, Dawood B, Donowitz M, Forman J, Pollitt RJ, Morgan NV, Tee L, Gissen P, Kahr WH, Knisely AS, Watson S, Chitayat D, Booth IW, Protheroe S, Murphy S, de Vries E, Kelly DA, Maher ER. 2010. Mutations in *TTC37* cause trichohepatoenteric syndrome (phenotypic diarrhea of infancy). Gastroenterology 138:2388-2398.

Kalisiak K, Kuliński TM, Tomecki R, Cysewski D, Pietras Z, Chlebowski A, Kowalska K, Dziembowski A. 2017. A short splicing isoform of HBS1L links the cytoplasmic exosome and SKI complexes in humans. Nucleic Acids Res 45(4):2068-2080.

Kammermeier J, Drury S, James CT, Dziubak R, Ocaka L, Elawad M, Beales P, Lench N, Uhlig HH, Bacchelli C, Shah N. 2014. Targeted gene panel sequencing in children with very early onset inflammatory bowel disease--evaluation and prospective analysis. J Med Genet 51:748-755.

Kammermeier J, Dziubak R, Pescarin M, Drury S, Godwin H, Reeve K, Chadokufa S, Huggett B, Sider S, James C, Acton N, Cernat E, Gasparetto M, Noble-Jamieson G, Kiparissi F, Elawad M, Beales PL, Sebire NJ, Gilmour K, Uhlig HH, Bacchelli C, Shah N. 2017.

This article is protected by copyright. All rights reserved.

Accepted Article

Phenotypic and Genotypic Characterisation of Inflammatory Bowel Disease Presenting Before the Age of 2 years. J Crohns Colitis 11:60-69.

Kinnear C, Glanzmann B, Banda E, Schlechter N, Durrheim G, Neethling A, Nel E, Schoeman M, Johnson G, van Helden PD, Hoal EG, Esser M, Urban M, Möller M. 2017. Exome sequencing identifies a novel *TTC37* mutation in the first reported case of Trichohepatoenteric syndrome (THE-S) in South Africa. BMC Med Genet 18(1):26.

Kopplin LJ, Igo RP Jr, Wang Y, Sivakumaran TA, Hagstrom SA, Peachey NS, Francis PJ, Klein ML, SanGiovanni JP, Chew EY, Pauer GJ, Sturgill GM, Joshi T, Tian L, Xi Q, Henning AK, Lee KE, Klein R, Klein BE, Iyengar SK. 2010. Genome-wide association identifies *SKIV2L* and MYRIP as protective factors for age-related macular degeneration. Genes Immun 11(8):609-621.

Kotecha UH, Movva S, Puri RD, Verma IC. 2012. Trichohepatoenteric syndrome: founder mutation in asian indians. Mol Syndromol 3:89-93.

Lebreton A, Séraphin B. 2008. Exosome-mediated quality control: substrate recruitment and molecular activity. Biochim Biophys Acta 1779(9):558-565.

Lee WI, Huang JL, Chen CC, Lin JL, Wu RC, Jaing TH, Ou LS. 2016a. Identifying Mutations of the Tetratricopeptide Repeat Domain 37 (*TTC37*) Gene in Infants With Intractable Diarrhea and a Comparison of Asian and Non-Asian Phenotype and Genotype: A Global Case-report Study of a Well-Defined Syndrome With Immunodeficiency. Medicine (Baltimore) 95(9):e2918.

Lee WS, Teo KM, Ng RT, Chong SY, Kee BP, Chua KH. 2016b. Novel mutations in *SKIV2L* and *TTC37* genes in Malaysian children with trichohepatoenteric syndrome. Gene 586:1-6.

This article is protected by copyright. All rights reserved.

Accepted Article

Mitchell P, Petfalski E, Shevchenko A, Mann M, Tollervey D. 1997. The exosome: a conserved eukaryotic RNA processing complex containing multiple 3'□5' exoribonucleases. Cell 91(4):457-466.

Monies DM, Rahbeeni Z, Abouelhoda M, Naim EA, Al-Younes B, Meyer BF, Al-Mehaidib A. 2015. Expanding phenotypic and allelic heterogeneity of tricho-hepato-enteric syndrome. J Pediatr Gastroenterol Nutr 60:352-356.

Morgan NV, Hartley JL, Setchell KD, Simpson MA, Brown R, Tee L, Kirkham S, Pasha S, Trembath RC, Maher ER, Gissen P, Kelly DA. 2013. A combination of mutations in AKR1D1 and *SKIV2L* in a family with severe infantile liver disease. Orphanet J Rare Dis 8:74.

Oz-Levi D, Weiss B, Lahad A, Greenberger S, Pode-Shakked B, Somech R, Olender T, Tatarsky P, Marek-Yagel D, Pras E, Anikster Y, Lancet D. 2015. Exome sequencing as a differential diagnosis tool: resolving mild trichohepatoenteric syndrome. Clin Genet 87:602-603.

Pegorado E and Hoffman EP. 2012. Limb-Girdle Muscular Dystrophy Overview. GeneReviews®[Internet].

Peter Sudbery. 2002. Human Molecular Genetics, 2d edition. Harlow, England: Prentice Hall - Cell and molecular biology in action series. 364p.

Richards S, Aziz N, Bale S, Bick D, Das S, Gastier-Foster J, Grody WW, Hegde M, Lyon E, Spector E, Voelkerding K, Rehm HL; ACMG Laboratory Quality Assurance Committee. 2015. Standards and guidelines for the interpretation of sequence variants: a joint consensus recommendation of the American College of Medical Genetics and Genomics and the Association for Molecular Pathology. Genet Med 17(5):405-424.

This article is protected by copyright. All rights reserved.

Accepted Article

Rider NL, Boisson B, Jyonouchi S, Hanson EP, Rosenzweig SD, Casanova JL, Orange JS. 2015. Novel *TTC37* Mutations in a Patient with Immunodeficiency without Diarrhea: Extending the Phenotype of Trichohepatoenteric Syndrome. Front Pediatr 3:2.

Rudnik-Schöneborn S, Senderek J, Jen JC, Houge G, Seeman P, Puchmajerová A, Graul-Neumann L, Seidel U, Korinthenberg R, Kirschner J, Seeger J, Ryan MM, Muntoni F, Steinlin M, Sztriha L, Colomer J, Hübner C, Brockmann K, Van Maldergem L, Schiff M, Holzinger A, Barth P, Reardon W, Yourshaw M, Nelson SF, Eggermann T, Zerres K. 2013. Pontocerebellar hypoplasia type 1: clinical spectrum and relevance of EXOSC3 mutations. Neurology 80(5):438-446.

Sankaran VG, Joshi M, Agrawal A, Schmitz-Abe K, Towne MC, Marinakis N, Markianos K, Berry GT, Agrawal PB. 2013. Rare complete loss of function provides insight into a pleiotropic genome-wide association study locus. Blood. 122(23):3845-3847.

Schaeffer D, Clark A, Klauer AA, Tsanova B, van Hoof A. 2010. Functions of the cytoplasmic exosome. Adv Exp Med Biol 702:79-90.

Stankler L, Lloyd D, Pollitt RJ, Gray ES, Thom H, Russell G. 1982. Unexplained diarrhoea and failure to thrive in 2 siblings with unusual facies and abnormal scalp hair shafts: a new syndrome. Arch Dis Child 57:212–216.

Vajsar J and Schachter H. 2006. Walker-Warburg syndrome. Orphanet J Rare Dis 1: 29.

Vanacova S, Stefl R. 2007. The exosome and RNA quality control in the nucleus. EMBO Rep 8(7):651-657.

Verloes A, Lombet J, Lambert Y, Hubert AF, Deprez M, Fridman V, Gosseye S, Rigo J, Sokal E. 1997. Tricho-hepato-enteric syndrome: further delineation of a distinct syndrome

This article is protected by copyright. All rights reserved.

Accepted Article

with neonatal hemochromatosis phenotype, intractable diarrhea, and hair anomalies. Am J Med Genet 68(4):391-395.

Wan J, Yourshaw M, Mamsa H, Rudnik-Schöneborn S, Menezes MP, Hong JE, Leong DW, Senderek J, Salman MS, Chitayat D, Seeman P, von Moers A, Graul-Neumann L, Kornberg AJ, Castro-Gago M, Sobrido MJ, Sanefuji M, Shieh PB, Salamon N, Kim RC, Vinters HV, Chen Z, Zerres K, Ryan MM, Nelson SF, Jen JC. 2012. Mutations in the RNA exosome component gene EXOSC3 cause pontocerebellar hypoplasia and spinal motor neuron degeneration. Nat Genet 44(6):704-708.

Zampieri S, Filocamo M, Pianta A, Lualdi S, Gort L, Coll MJ, Sinnott R, Geberhiwot T, Bembi B, Dardis A. 2016. SMPD1 Mutation Update: Database and Comprehensive Analysis of Published and Novel Variants. Hum Mutat 37(2):139-147.

Zheng B, Pan J, Jin Y, Wang C, Liu Z. 2016. Targeted next-generation sequencing identification of a novel missense mutation of the *SKIV2L* gene in a patient with trichohepatoenteric syndrome. Mol Med Rep 14:2107-2110.

Zhu B, Mandal SS, Pham AD, Zheng Y, Erdjument-Bromage H, Batra SK, Tempst P, Reinberg D. 2005. The human PAF complex coordinates transcription with events downstream of RNA synthesis. Genes Dev 19(14):1668-16673.

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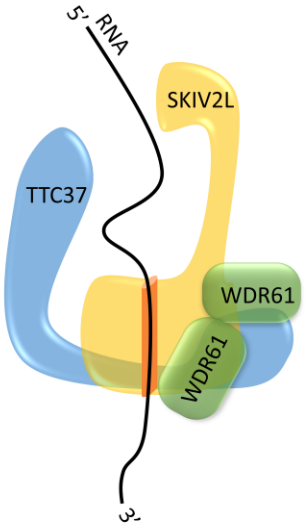
Figure legends

Figure 1. Model prediction of the SKI complex and names of orthologues among different species. The SKI complex is thought to act as a guide for RNA capture. Molecular structure has been modeled from crystallography data in yeast. Adapted from [Halbach et al., 2013].



S. cerevisiae	*A. thaliana*	*D. Melanogaster*	*H. sapiens*
SKI2	At3g46960	Twister	SKIV2L
SKI3	At1g76630	CG8777	TTC37
SKI8	At4g29830	CG3909	WDR61

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Figure 2. *SKIV2L* and *TTC37* mutation spectrum, global analysis. **A:** geographical distribution of mutations around the world by gene. Light color, previously reported mutations. Plain color, French cohort. **B:** mutational status of patients by gene. C: types of mutations by gene.

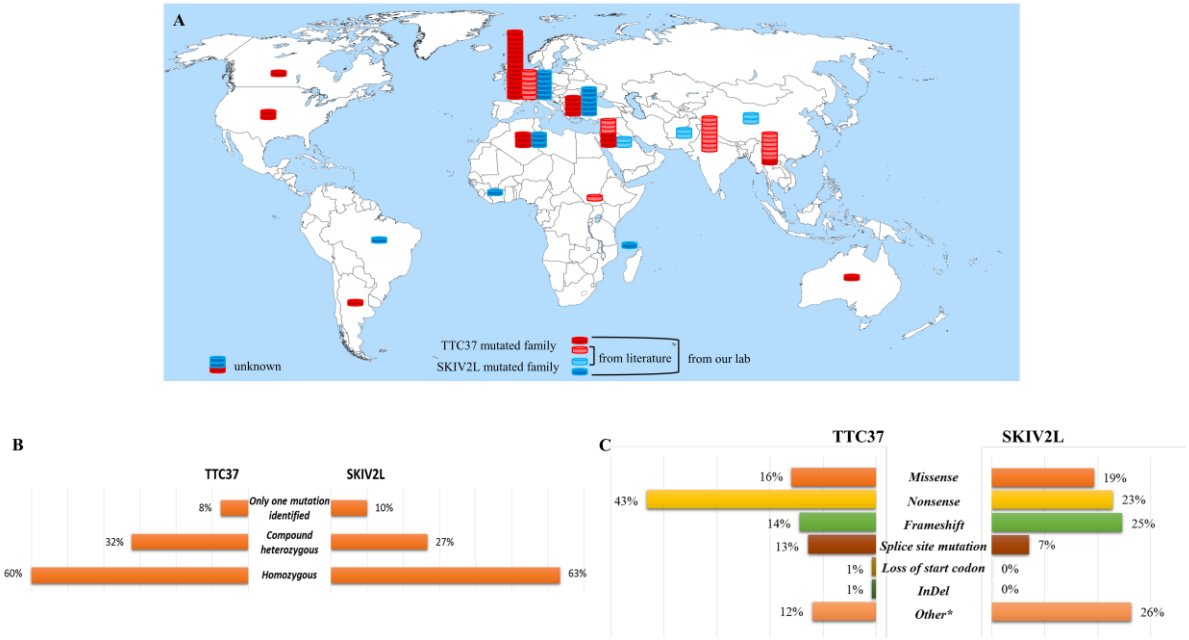

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Figure 3. *SKIV2L* and *TTC37* mutation spectrum, molecular analysis. **A:** schematic representation (not to scale) of human *SKIV2L* gene locus and structure. Exons are represented as boxes, introns as tubes. Exon numbering is indicated, non-colored boxes are untranslated regions. Below are shown both localization and type of mutation. Lower panel is the schematic representation of corresponding protein, with functional domains indicated. Numbers correspond to amino acids. **B:** same representation for human *TTC37* gene and protein. Note that the TRP domains of *TTC37* are only predictions from the yeast homolog.

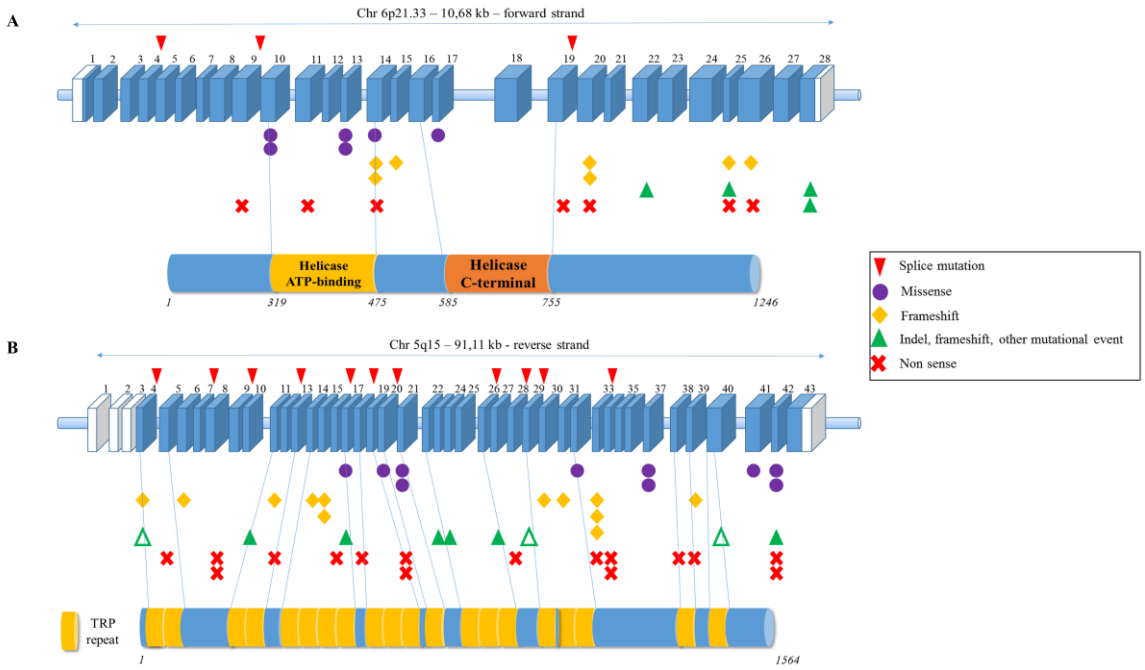

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Figure 4. Phenotype-genotype correlation analysis. **A:** distribution of clinical signs by gene (only if found and only for general signs). This scheme comprises the whole cohort of both published and non-published patients. **B:** schematic representation of birth height in percentiles. Groups of patients (histograms) correspond to each standard deviation around the 50th percentile. Normal distribution of percentiles is drawn on background. **C:** same representation as B for birth weight. Please note that patients mutated for either gene are pooled together to ensure a minimal number of data. These two diagrams only concern the Lab's cohort, as these parameters are usually not available in literature.

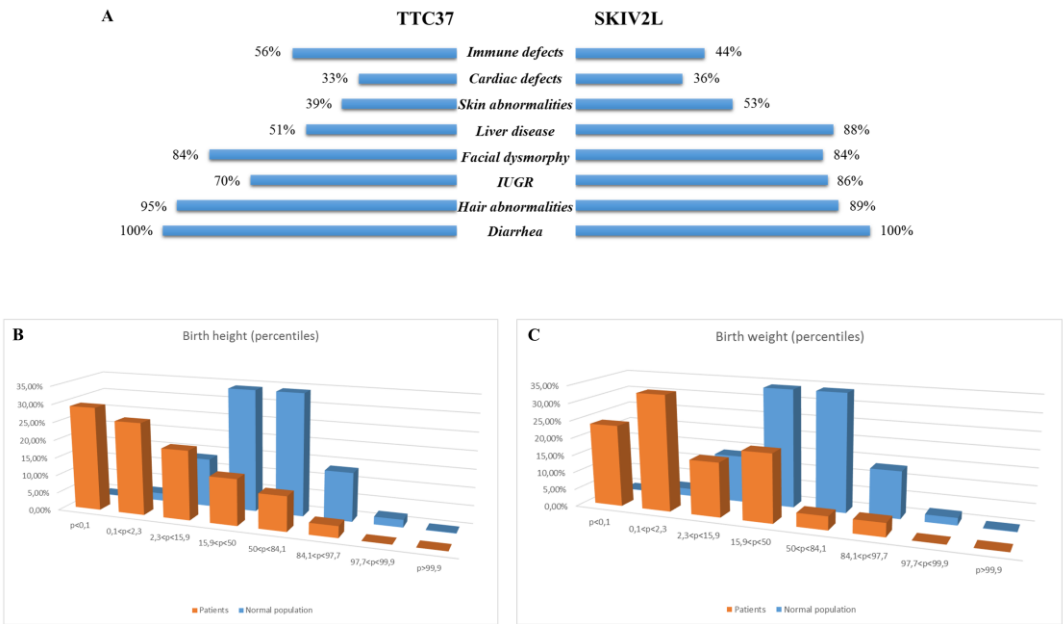

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Table 1. General statistics on patients and mutations

	Number of patients	Number of families	Number of mutations	Mutated alleles count	Number of novel mutations[a]	Number of patients from the French cohort[b]	Number of homozygous	Number of compound heterozygous	Number of patients with only one mutation identified[c]	Males/ Females
TTC37	65	58	58	125	24	38	39	21	5	30/35
SKIV2L	31	27	28	59	13	21	19	9	3	15/16
Total	**96**	**85**	**86**	**184**	**37**	**59**	**58**	**30**	**8**	**45/51**

a: unpreviously reported

b: both published and new ones

c: in the diagnostic context

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Table 2. Summary of variants in TTC37

Exon / Intron	Nucleotide change (NM_014639.3)	Predicted amino acid change	Nature of effect	Number of alleles	Number of patients (Families)	gnomAD frequency	Genomic coordinates (GRCh37/hg19)	References	Remarks
Exon 4	c.3 G>A	p.Met1Ile, p.0?	Loss of start codon	1	1	-	chr5:94882852	This study	
	c.52 delA	p.Arg18Glu fs*9	Frameshift	1	1	-	chr5:94882803	This study	
Intron 4	c.91-1 G>C		Splice mutation	2	1	-	chr5:94879032	This study	
Exon 5	c.154 G>T	p.Glu52*	Nonsense	1	1	-	chr5:94878968	Chong & al 2015	
Exon 6	c.287_291 delTGCCT	p.Leu96Trp fs*11	Frameshift	4	2 (1)	-	chr5:94877554-94877558	Fabre & al. 2011	
Intron 7	c.402+2 T>G		Splice mutation	1	1	-	chr5:94877007	This study	
Exon 8	c.409 C>T	p.Arg137*	Nonsense	2	1	8.128e-6	chr5:94876528	This study	rs776321294
	c.439 C>T	p.Gln147*	Nonsense	1	1	-	chr5:94876498	Hartley & al, 2010	
Intron 9	c.643-2 A>G		Splice mutation	2	1	-	chr5:94872631	This study	
Exon 10	c.751 G>A	(p.Gly251Arg) p.Phe215Glu fs*14	Predicted as a missense, but generates an exon skipping	2	1	4.064e-6	chr5:94872521	Hartley & al, 2010	rs763816083
Exon 11	c.786 T>A	p.Cys262*	Nonsense	2	1	-	chr5:94865899	This study	c.786 T>C (synonymous) reported once in gnomAD
	c.828 delT	p.Ile276Met fs*3	Frameshift	1	1	-	chr5:94865857	This study	
Intr	c.994+1 G>A		Splice	1	1	4.069	chr5:94864	This study	rs771917

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on 12			mutation			e-6	704		451
Exon 14	c.1168 delG	p.Val390Phefs*30	Frameshift	1	1	4.067e-6	chr5:94861344	Fabre & al. 2011	Not found in dbSNP
Exon 15	c.1300_1301 delAA	p.Lys434Glyfs*14	Frameshift	1	1	4.068e-6	chr5:94861126-94861127	Hartley & al, 2010	Nomenclature corrected, not found in dbSNP
	c.1307_1308 delAT	p.Tyr436Serfs*12	Frameshift	1	1	-	chr5:94861119-94861120	This study	
Exon 16	c.1374 C>G	p.Tyr458*	Nonsense	1	1	2.44 e-5	chr5:94860247	This study	rs373796916
Intron 16	c.1453-1 G>C		Splice mutation	1	1	-	chr5:94859539	This study	
Exon 17	c.1499 G>T	p.Gly500Val	Missense	2	2 (1)	2.846e-5	chr5:94859492	This study	rs138144509
Intron 17	c.1632+1 delG	p.Glu545Phefs*40	Splice mutation with exon skipping	2	1	-	chr5:94859358	Hartley & al, 2010	
Exon 18	c.1708 C>T	p.Arg570*	Nonsense	3	2	8.131e-6	chr5:94858954	This study	rs768215813
Intron 18	c.1757+2 T>G		Splice mutation	1	1	7.218e-6	chr5:94858904	This study	rs758588837
Intron 20	c.2114+5 G>A		Splice mutation	2	1	5.785e-5	chr5:94856415	This study	rs775874166
Exon 20	c.2018 G>A	p.Gly673Asp	Missense	1	1	8.154e-6	chr5:94856516	Kammermeier & al, 2014	rs769075304
Exon 21	c.2128 C>T	p.Arg710*	Nonsense	1	1	-	chr5:94853013	Rider & al, 2015	
	c.2161 G>A	p.Gly721Arg	Missense	2	2 (1)	8.159e-6	chr5:94852980	This study	Not found in dbSNP
	c.2251 C>T	p.Gln751*	Nonsense	1	1	-	chr5:94852890	Hartley & al, 2010	
	c.2282 T>C	p.Leu761Pro	Missense	2	1	-	chr5:94852859	OZ-Levy & al, 2015	Bad labelling of coding strand in the ref.
Intron 23	c.2515+1 G>C	p.Cys813Valfs*56	Splice mutation with exon skipping	2	1	-	chr5:94852375	Fabre & al. 2011	
Intron 24	c.2578-7_2578-3 del	p.(Asn860_Glu878del)	Splice mutation with	1	1	3.981e-5	chr5:94852116-94852120	Fabre & al. 2011	rs746874042

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			exon skipping						
Intron 27	c.2779-2 A>G	p.Glu974Glyfs*19	Splice mutation with exon skipping	4	2 (2)	-	chr5:94848324	Hartley & al, 2010	
Exon 28	c.2808 G>A	p.Trp936*	Nonsense	15	8 (8)	6.096e-5	chr5:94848293	Hartley & al, 2010, Kammermeier & al, 2014, Kotecha & al, 2012, this study	rs534237033
Intron 28	c.2921-2 A>G		Splice mutation	4	2	6.115e-5	chr5:94845393	Bozzetti & al, 2013, this study	
Exon 29	c.2990_3031delinsC		Indel	1	1	-	chr5:94845322-94842700	This study	
Intron 29	c.3015-1 G>A		Splice mutation	1	1	-	chr5:94842716	Fabre & al. 2011	
Exon 30	c.3051_3052insA	p.Thr1018Asnfs*18	Frameshift	1	1	-	chr5:94842678	This study	
Exon 31	c.3185_3201dup	p.Lys1068Serfs*2	Frameshift	1	1	-	chr5:94839550-94839533	This study	
Exon 32	c.3230 C>A	p.Ala1077Asp	Missense	1	1	-	chr5:94838695	Fabre & al. 2011	c.3230 C>T reported once in gnomAD
Exon 33	c.3364 delA	p.Thr1122Glnfs*10	Frameshift	1	1	-	chr5:94834273	This study	Substitution reported once in gnomAD
	c.3426 dupA	p.Ala1143 fs*4	Frameshift	3	2 (2)	-	chr5:94834211	Lee & al. 2016a, Lee & al. 2016b	Nomenclature corrected from the second paper
	c.3464_3465delAA	p.Lys1155 fs*2	Frameshift	2	1	5.05 e-5	chr5:94834172-94834171	Lee & al. 2016a	Nomenclature corrected, rs749337189
	c.3507 T>G	p.Tyr1169*	Nonsense	3	3 (3)	5.411e-5	chr5:94834130	Chong & al 2015, Lee & al.	rs774353983

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								2016a	
Intron 33	c.3564-2 A>G		Splice mutation	1	1	-	chr5:94833194	Fabre & al. 2011	
Exon 34	c.3601 C>T	p.Arg1201*	Nonsense	1	1	-	chr5:94833155	Chong & al 2015	Nomenclature corrected
	c.3613 C>T	p.Gln1205*	Nonsense	1	1	-	chr5:94833143	This study	
Exon 37	c.3808 C>G	p.Pro1270Ala	Missense	2	1	0.0006111	chr5:94826735	Fabre & al. 2011	rs146627706
	c.3847 G>A	p.Asp1283Asn	Missense	2	2	-	chr5:94826696	Hartley & al, 2010	
Exon 38	c.3960 C>A	p.Tyr1320*	Nonsense	2	1	-	chr5:94820521	Fabre & al. 2011	c.3960 C>T (synonymous) reported five times
Exon 39	c.4059 delA	p.Ser1355Valfs*12	Frameshift	1	1	-	chr5:94818330	Busoni & al. 2016	
	c.4102 C>T	p.Gln1368*	Nonsense	2	1	4.063e-6	chr5:94818287	Monies & al, 2015	rs763823697
Exon 40	c.4337_4338insCTA	p.Ala1447*	Insertion	1	1	-	chr5:94814021	Rider & al, 2015	
Exon 41	c.4454 T>G	p.Leu1485Arg	Missense	1	1	-	chr5:94805526	Fabre & al. 2011	
Exon 42	c.4507 C>T	p.Arg1503Cys	Missense	4	2 (1)	5.782e-5	chr5:94803683	Kinnear & al. 2017	One family, rs200067423
	c.4514 T>C	p.Leu1505Ser	Missense	3	3 (3)	0.0001048	chr5:94803676	Hartley & al, 2010, this study	Nomenclature corrected, rs376720108
	c.4572 G>A	p.Trp1524*	Nonsense	17	9 (6)	4.064e-6	chr5:94803618	Fabre & al. 2017a, this study	6 turkish families, rs771811650
	c.4575 C>G	p.Tyr1525*	Nonsense	1	1	-	chr5:94803615	Busoni & al. 2016	c.4575 C>T (synonymous) reported twice
Intron 42	c.4620+1 G>C	p.Trp1524_1564 DelIns61	Splice mutation with indel	1	1	4.067e-6	chr5:94803569	Fabre & al. 2011	rs370373017

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Table 3. Summary of variants in SKIV2L

Exon / Intron	Nucleotide change (NM_006929.4)	Predicted amino acid change	Nature of effect	Number of alleles	Number of patients (Families)	gnomAD frequency	Genomic coordinates (GRCh37/hg19)	References	Remarks
Intron 4	c.355-2 A>C		Splice mutation	2	1	-	chr6:31928207	Kammermeier & al, 2014	
Exon 9	c.848 G>A	p.Trp283*	Nonsense	1	1	1.653 e-5	chr6:31929382	Fabre & al. 2012	rs751026211
Intron 9	c.919-1 G>A		Splice mutation	1	1	4.148 e-6	chr6:31929685	This study	c.919-1 G>C reported at same frequency
Exon 10	c.994 G>C	p.Ala332Pro	Missense	1	1	-	chr6:31929761	This study	
	c.1022 T>G	p.Val341Gly	Missense	1	1	-	chr6:31929789	Fabre & al. 2012	
Exon 11	c.1120 C>T	p.Arg374*	Nonsense	1	1	2.526 e-5	chr6:31930271	Zheng & al. 2016	rs200818962
Exon 13	c.1312 G>A	p.Glu438Lys	Missense	2	1	-	chr6:31930777	This study	
	c.1396 T>G	p.Trp466Gly	Missense	2	1	-	chr6:31930861	This study	
Exon 14	c.1434 delT	p.Ser479Ala fs*3	Frameshift	4	2 (2)	-	chr6:31931220	Fabre & al. 2012	
	c.1447 C>T	p.Arg483Cys	Missense	2	1	1.093 e-5	chr6:31931233	This study	rs779670171
	c.1452 delC	p.Pro484 fs*46	Frameshift	1	1	-	chr6:31931238	This study	c.1452 C>T (synonymous)reported 3 times
	c.1528 C>T	p.Arg510*	Nonsense	2	2 (1)	1.758 e-5	chr6:31931314	This study	rs768131466
Exon 15	c.1635_1636 insA	p.Gly546Arg fs*35	Frameshift	2	1	-	chr6:31931498-31931499	Fabre & al. 2012	
Exon 17	c.1891 G>A	p.Gly631Ser	Missense	3	3 (3)	-	chr6:31932039	Lee & al. 2016b, Zheng & al. 2016	
Intron 18	c.2203-1 G>C		Splice mutation	1	1	4.066 e-6	chr6:31934485	Bick & al. 2016	c.2203-1 G>A reported at same frequency
Exon 19	c.2266 C>T	p.Arg756*	Nonsense	1	1	1.218 e-5	chr6:31934549	Fabre & al. 2012	rs372977009
Intron 19	c.2341-2 A>G		Splice mutation	1	1	1.24 e-5	chr6:31934779	This study	rs781763471

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Exon 20	c.2442 G>A	p.Trp814*	Nonsense	1	1	4.162 e-6	chr6:31934882	Fabre & al. 2012	rs148221996 c.2442 G>C and c.2442 G>T (missenses) also reported under this rs
	c.2498_2499 delTG	p.Val833Glu fs*45	Frameshift	1	1	3.232 e-5	chr6:31935069-31935070	This study	Not found in dbSNP
	c.2572 delG	p.Val858*	Frameshift	2	1	-	chr6:31935143	Fabre & al. 2012	
Exon 22	c.2662_2664 delAGG	p.Arg888del	In frame deletion	2	1	-	chr6:31935570-31935572	Fabre & al. 2012	Reported as c.2662_2663del (p.Arg888Gly fs*12)
Exon 25	c.3101_3103 delAGA	p.Gln1034del	In frame deletion	1	1	-	chr6:31936494-31936496	This study	
	c.3112_3140 del	p.Glu1038 fs*7	Frameshift	2	1	-	chr6:31936505-31936533	This study	
	c.3172 C>T	p.Arg1058*	Nonsense	2	1	4.156 e-6	chr6:31936565	This study	rs759428204,c.3172 C>A reported at same frequency
Exon 26	c.3187 C>T	p.Arg1063*	Nonsense	6	5 (5)	0.0001179	chr6:31936654	Lee & al. 2016b, Bick & al. 2016, this study	rs138923214
	c.3391 delC	p.Leu1131Ser fs*5	Frameshift	2	1	-	chr6:31936858	Morgan & al, 2013	
Exon 28	c.3559_3579 del	p.Gly1187_Glu1193del	In frame deletion	8	4 (2)	-	chr6:31937310-31937330	Monies & al, 2015	
	c.3561_3581 del	p.Ser1189_Leu1195del	In frame deletion	4	2 (2)	-	chr6:31937312-31937332	This study	

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Table 4. Numeric clinical data of the "Lab Cohort"

		Sample size	Minimal	Maximal	Median	Mean value	Standard deviation	p-value (t-test)
Height (percentiles)	SKIV2L	9/21	0.00	13.58	0.10	2.20	4.46	
	TTC37	22/38	0.00	95.21	3.65	19.89	29.25	0.01108
Weight (percentiles)	SKIV2L	18/21	0.00	43.07	0.62	6.09	12.85	
	TTC37	33/38	0.00	95.46	2.92	17.24	26.32	0.04788
Birth (week)	SKIV2L	18/21	32.00	40.00	37.29	36.82	2.43	
	TTC37	35/38	30.00	41.00	36.57	36.34	2.54	0.50877
Cranial perimeter (percentiles)	SKIV2L	4/21	0.00	5.35	0.68	1.68	2.49	
	TTC37	9/38	0.01	85.54	7.00	25.44	31.44	0.053556
Parenteral nutrition (day of onset)	SKIV2L	12/21	1.00	579.50	105.50	159.54	170.39	
	TTC37	25/38	1.00	518.00	90.00	119.43	119.87	0.47386

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Table 5. Clinical study of the "Lab Cohort"

		Global (n=59)	TTC37 mutated (n=38)	SKIV2L mutated (n=21)
		No. of positive/No. of patients	No. of positive/No. of patients	No. of positive/No. of patients
Dysmorphism		42/53 (79%)	29/35 (83%)	13/18 (72%)
	Facial dysmorphism	38/47 (81%)	27/33 (82%)	11/14 (79%)
	Hypertelorism	30/44 (68%)	18/29 (62%)	12/15 (80%)
	Broad forehead	32/46 (70%)	23/34 (68%)	9/12 (75%)
Hair abnormalities		52/54 (96%)	33/34 (97%)	19/20 (95%)
	Wooly hair	40/46 (87%)	27/31 (81%)	13/15 (87%)
	Trichorrhexis nodosa	28/39 (72%)	18/26 (69%)	10/13 (77%)
	Easily removable	32/37 (87%)	22/26 (85%)	10/11 (91%)
Liver abnormalities		30/52 (58%)	18/34 (53%)	12/18 (67%)
Primary	Liver disease	15/44 (34%)	7/29 (24%)	8/15 (53%)
	Cholestasis	9/40 (23%)	3/25 (12%)	6/15 (40%)
	High transaminases levels	11/40 (28%)	6/26 (23%)	5/14 (36%)
	Cirrhosis	4/38 (11%)	1/25 (4%)	3/13 (23%)
	Hepatomegaly	10/37 (27%)	4/23 (17%)	6/14 (43%)
Secondary (to PN)	Liver disease	17/34 (50%)	13/25 (52%)	4/9 (44%)
	Cholestasis	7/33 (21%)	5/24 (21%)	2/9 (22%)
	High transaminases levels	12/35 (34%)	8/23 (35%)	4/12 (33%)
Dysimmunity		34/47 (72%)	24/33 (73%)	10/14 (71%)
	Low Ig levels	23/43 (53%)	16/30 (53%)	7/13 (54%)
	Functionnel deficiency	25/36 (69%)	18/25 (72%)	7/11 (64%)
	Macrophage activation syndrome	3/26 (12%)	2/17 (12%)	1/9 (11%)
	Platelet anomalies	8/40 (20%)	4/28 (14%)	4/12 (33%)
Dermatological anomalies		25/49 (51%)	13/32 (41%)	12/17 (71%)
	Café au lait spots	15/31 (48%)	8/16 (50%)	7/15 (47%)
Diarrhea		55/55 (100%)	36/36 (100%)	19/19 (100%)
Cardiac defects		9/45 (20%)	4/32 (13%)	5/13 (38%)
Intellectual disability		17/39 (44%)	10/25 (40%)	7/14 (50%)
Hypothyroïdism		3/28 (11%)	1/20 (5%)	2/8 (25%)
Parenteral nutrition		44/52 (85%)	25/32 (88%)	16/20 (80%)
Ig supplementation		17/39 (44%)	13/27 (48%)	4/12 (33%)

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